Cambridge Assessment International Education

Cambridge O Level

CHEMISTRY **5070/22**

Paper 2 Theory **May/June 2021**

MARK SCHEME

Maximum Mark: 75

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2021 series for most Cambridge IGCSE™, Cambridge International A and AS Level components and some Cambridge O Level components.

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Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

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GENERIC MARKING PRINCIPLE 5: Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).
GENERIC MARKING PRINCIPLE 6: Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1	Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
2	The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
3	Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
4	The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
5	‘List rule’ guidance For questions that require ***n*** responses (e.g. State **two** reasons …): • The response should be read as continuous prose, even when numbered answer spaces are provided. • Any response marked *ignore* in the mark scheme should not count towards ***n***. • Incorrect responses should not be awarded credit but will still count towards ***n***. • Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response. • Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

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6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

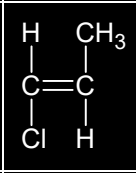
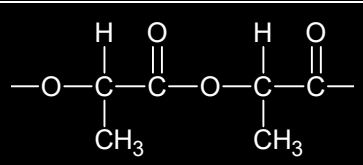
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Question	Answer	Marks
1(a)(i)	C	**1**
1(a)(ii)	A	**1**
1(a)(iii)	B	**1**
1(a)(iv)	D	**1**
1(a)(v)	H	**1**
1(b)	C and F	**1**

Question	Answer	Marks
2(a)	21 (%)	**1**
2(b)	steel making / welding / cutting metals / oxygen tents in hospitals / breathing apparatus / fuel cells	**1**
2(c)(i)	same electronic structure / same electronic configuration	**1**
2(c)(ii)	polonium-210 has one more neutron / polonium-209 has one less neutron / polonium-209 has 125 neutrons and polonium-210 has 126 neutrons	**1**
2(d)	percentage of selenium = 47.6 (1) mole ratio Se : O : C*l* is 47.6 / 79 : 9.6 / 16 : 42.8 / 35.5 **OR** 0.60 : 0.60 : 1.21 (1) empirical formula SeOCl_2 (1)	**3**
2(e)(i)	volume decreases **AND** particles move slower / particles less spread out / particles closer together	**1**

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Question	Answer	Marks
2(e)(ii)	volume increases **AND** particles move further apart / particles more spread out / distance between particles increases	**1**
2(e)(iii)	mol = 11.5 / 24 **OR** 0.479(117) (1) mass = 15 (g) (1)	**2**

Question	Answer	Marks
3(a)(i)	addition	**1**
3(a)(ii)		**1**
3(a)(iii)	makes poisonous gases / makes poisonous fumes	**1**
3(b)(i)	can be decomposed naturally / can be broken down by bacterial action	**1**
3(b)(ii)	 one ester linkage drawn correctly (1) structure with at least two repeat units and extension bonds at both ends (1)	**2**
3(b)(iii)	condensation polymerisation / a small molecule is also made	**1**
3(b)(iv)	oxidised	**1**
3(b)(v)	$H^+ + OH^- \rightarrow H_2O$	**1**

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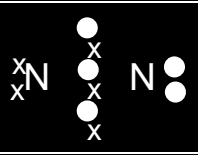
Question	Answer	Marks
3(b)(vi)	hydrogen	1

Question	Answer	Marks
4(a)	**Any two from:** soluble in water (1) insoluble in organic solvents (1) high melting point / solid at room temperature (1) white (1)	2
4(b)	zinc loses electrons and bromine gains electrons (1) reference to (transfer of) two electrons (1)	2
4(c)(i)	magnesium (atom) loses electrons	1
4(c)(ii)	Zn^{2+} (ion) gains electrons / zinc ions gain electrons	1
4(d)(i)	sodium carbonate / potassium carbonate / ammonium carbonate	1
4(d)(ii)	zinc oxide **AND** carbon dioxide	1

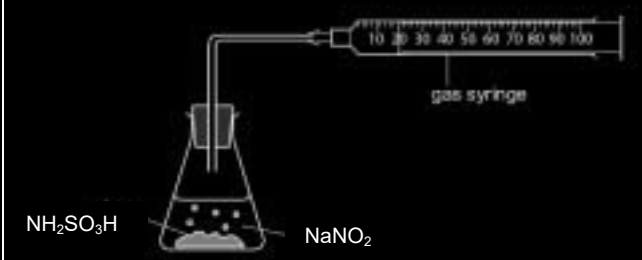
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Question	Answer	Marks
5(a)	**Any two from:** crude oil heated / crude oil vapourised (1) fractional distillation / use of fractionating column / use of fractionating tower (1) separated because of different boiling points (1) fractions move up the column at different rates / fractions condense at different heights in the column (1)	**2**
5(b)(i)	fits the formula C_nH_{2n+2} / it is $C_{11}H_{(11 \times 2) + 2}$	**1**
5(b)(ii)	$C_{11}H_{24} \rightarrow C_2H_4 + C_9H_{20}$ / $C_{11}H_{24} \rightarrow 2C_2H_4 + C_7H_{16}$ / $C_{11}H_{24} \rightarrow 3C_2H_4 + C_5H_{12}$ / $C_{11}H_{24} \rightarrow 4C_2H_4 + C_3H_8$ / $C_{11}H_{24} \rightarrow 5C_2H_4 + CH_4$	**1**
5(c)	(reaction with) nitrogen (1) **AND** **Any two conditions from:** Fe (catalyst) (1) 400 to 500 °C (1) 200 to 400 atmosphere (pressure) (1)	**3**
5(d)	making margarine / fuel cells / fuel	**1**
5(e)(i)	water / steam	**1**
5(e)(ii)	heat / high temperature / catalyst	**1**

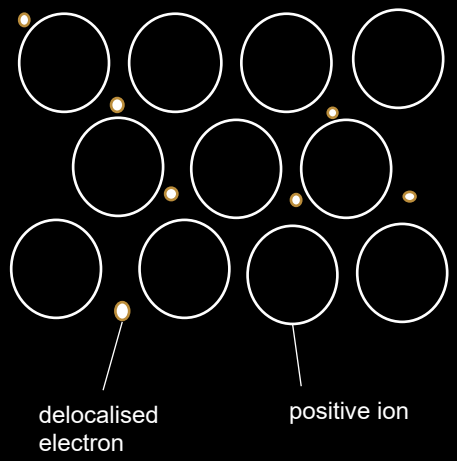
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Question	Answer	Marks
6(a)	acid rain	**1**
6(b)	calcium carbonate	**1**
6(c)(i)	$N_2 + O_2 \rightarrow 2NO$	**1**
6(c)(ii)		**1**
6(c)(iii)	particles move faster / particles have more kinetic energy (1) more successful collisions / more (particles) have energy above the activation energy / more effective collisions (1)	**2**
6(d)(i)	$2NO_2 + 2NaOH \rightarrow NaNO_2 + NaNO_3 + H_2O$ formula for $NaNO_2$ (1) balancing (1)	**2**
6(d)(ii)	strong acid completely dissociates / completely ionises (1) weak acid incompletely dissociates / incompletely ionises / partially dissociates / partially ionises (1)	**2**

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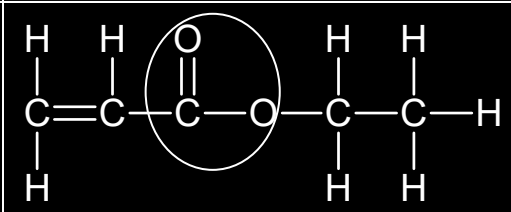
Question	Answer	Marks
7(a)	moles of $NaNO_2$ = 0.00300 (1) volume = 0.072 (dm^3) (1)	**2**
7(b)	 correct method of collection including arrangement of apparatus (1) workable apparatus e.g. no gas leaks (1)	**2**
7(c)	increases (1) more crowded particles / more particles per unit volume / smaller space between particles (1) collision frequency increases / more collisions per second / increase rate of collisions (1)	**3**
7(d)	$2NH_2SO_3H + MgCO_3 \rightarrow Mg(NH_2SO_3)_2 + H_2O + CO_2$ CO_2 as a product (1) correct formula for magnesium sulfamate (1) balanced equation (1)	**3**

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Question	Answer	Marks
8(a)(i)	six	**1**
8(a)(ii)	four	**1**
8(b)	 closely packed positive ions in regular arrangement (1) delocalised electrons / sea of electrons (1) strong attraction between the (positive) ions and the electrons (1)	**3**
8(c)	**Any two for one mark:** (1) good electrical conductor good thermal conductor shiny / lustrous malleable ductile/ can be made into wires	**1**
8(d)	ethanoic acid and lead(II) oxide or lead(II) hydroxide	**1**

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Question	Answer	Marks
8(e)	$Pb^{2+}(aq) + 2I^{-}(aq) \rightarrow PbI_2(s)$ correct formulae for reactants and products and balanced (1) state symbols dependent on correct formulae (1)	**2**
8(f)	ions cannot move / no mobile ions	**1**

Question	Answer	Marks
9(a)		**1**
9(b)	(reacts with the) carbon-carbon double bond	**1**
9(c)(i)	ethanol	**1**
9(c)(ii)	no effect	**1**
9(c)(iii)	moves to the right / moves to the side of the ester (1) uses up the extra alcohol / to reduce concentration of alcohol (1)	**2**
9(d)(i)	mol of carboxylic acid = 10.8 / 72 **OR** 0.15 mol (1) relative molecular mass of ester = 100 (1) mass of ester = 0.15 × 100 (1)	**3**
9(d)(ii)	63 (%)	**1**

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